
Analysis of Plastic Waste Management: Utilization, Issues & Solution

¹Gunjan, ²Ruchi Bharti, ³Renu Sharma

^{1,2,3}Department of Chemistry, University Institute of Science, Chandigarh University,
Gharuan, Mohali, Punjab.

²E-Mail: ruchi.uis@cumail.in

Abstract

Last 20 decades, plastics have been used for many purposes and become an important part of our daily life. In many developing countries, people use the bags, bottles, utensils, furniture made up of plastics and for many more domestic purposes without considering the fact that it will create a lot of waste which is impossible to manage. Due to this fact, it has recently become a common and serious issue worldwide. It harms not only the environment but also the human beings. Along with this, tons of plastics exist in oceans which causes loss of marine animals every year. Looking at this serious issue, there is a need to manage the plastic wastes by different methodologies like recycling and reusing, landfill, incineration, gasification and hydrogenation etc. In this review paper, we are going to look towards the current procedures of use of plastic waste in current situation as well as to look for all the possibilities to minimize these wastes.

Keywords: Plastic Waste, Recycling, Plastic Roads, Incineration

1 Introduction

In our everyday life, plastic items are used in large amounts. In several cities around the world, there is quick populace development, urbanization and mechanical development which causes a huge waste administration issue

Journal of Green Engineering, Vol. 10_10, 8524-8541.

© 2020 Alpha Publishers. All rights reserved

[1], [2]. From the past century, plastic production has increased from an unimportant 1.3 million tons in 1950 to >322 million tons in 2015. In today's life, plastic is everywhere. Plastic is collected from both the rural and urban areas [3], [4]. Plastic is toxic in nature which is found in different forms. We can solve the problem of plastic waste by two main possibilities: First possibility is to substitute the plastic with other materials and second is to collect the plastic waste and recycle it. Main methods of recycling the plastic waste are: Chemically, Thermally, and by material. Thermal Recycling Utilizes just a little part of the vitality put away in the waste. One of its major advantage that there is no need to separate from other flammable waste, it is not being a priority when there is plastic waste energy substitutes oil, gas or coal in production. Chemical Recycling utilizes a large part of the stored energy than the thermal recycling [5], [6]. It is based on the depolarisation of recycled waste monomers and these monomers are used for the preparation of new virgin polymers. This method is very difficult to apply on the mixed plastic waste because different methods of depolarization are used for different polymers. The method of depolarization and chemical nature of plastic tells the efficiency of chemical recycling of plastic waste. Material Recycling of plastic waste uses the larger part of the stored energy. It is mainly used for a single type of industrial waste, where only reprocessing and additional stabilization is required. The recycled materials are used only for limited applications; therefore, low prices are obtained in the market. Figure 1 shows Total Quantity of Plastic Waste Generated Per Day.

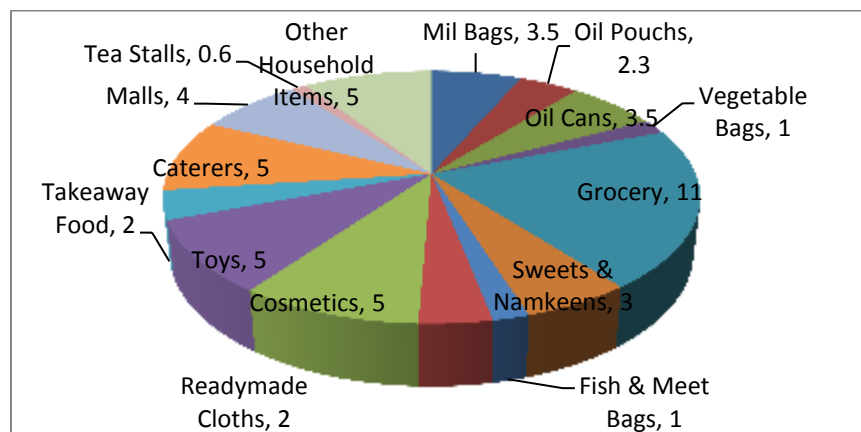


Figure 1 Total Quantity of Plastic Waste Generated Per Day. [3]

2 Utilization of Plastic Waste within the Construction of Road

A standout amongst other strategy to utilise or reuse the plastic waste is construction of roads with that plastic waste. Road construction from plastic waste isn't new [7], [8]. Utilizing polymers changed bitumen to gain higher asphalt pavement generally speaking execution has been resolved for a long term. The main material of this method is 'Bitumen and Plastic'.



Figure 2 Plastic bitumen road [7]

Bitumen: It acts as aggregates in bitumen mixes. 85% of all bitumen is used as a binder in asphalt for foot paths, runways, roads and parking lots [9], [10]. By the use of volatile diluent, sometimes the viscosity of bitumen is reduced, this material is known as cutback. Figure 2 shows Plastic bitumen road. Table 1 provides detail about Bitumen Extraction at Certain Interval.

Table 1 Bitumen Extraction at Certain Interval

Plastic Content (% By Weight)	Bitumen Extracted After 5 Min %	Bitumen Extracted After 10 Min %	Bitumen Extracted After 15 Min %
0.0	95.0	97.0	98.0
0.5	65.0	87.9	92.3
0.75	62.7	85.3	90.7
1.0	60.6	76.2	82.8

Plastic: Three types of plastics are used: Thermosets, Elastomers, Thermoplastic. Plastic wastes used are: a) Soft foams with any thickness i.e.,

PE&PP, b) Hard foams with any thickness i.e., PS, c) Laminated plastic and films (i.e., carry bags, cups) up to the 60 μ thickness. The plastic wastes that are not used are PVC and Flux sheets . Table 2 provides detail about Plastic Consumption in India. Table 3 provides detail about Polymer Demand in India (Million Tonnes). Table 4 provides detail about Plastic Waste Consumption.

Table 2 Plastic Consumption in India

S.No.	Year	Consumption (Tonnes)
1	1996	61000
2	2001	400000
3	2006	700000
4	2011	135000
5	2013	174000

Table 3 Polymer Demand in India (Million Tonnes)

Types of Polymer	1995-96	2001-02	2006-07	2010-11
PE	0.83	1.83	3.27	7.12
PP	0.34	0.88	1.79	3.88
PVC	0.49	0.87	1.29	2.87
PET	0.03	0.14	0.29	0.75
TOTAL	1.69	3.72	6.64	14.62

Table 4 Plastic Waste Consumption

S.No.	Description	World	India
1	Per Capita Per Year Consumption of Plastic (Kg)	24-28	12-16
2	Recycling (%)	25	60
3	Plastic in Solid Waste (%)	7	9

Now the question arises why we use plastic? So the reason is plastic is durable and resisted with corrosion, it is light weight, it is maintenance free, and it has a longer life. Figure 3 shows Flow Chart for Polymers

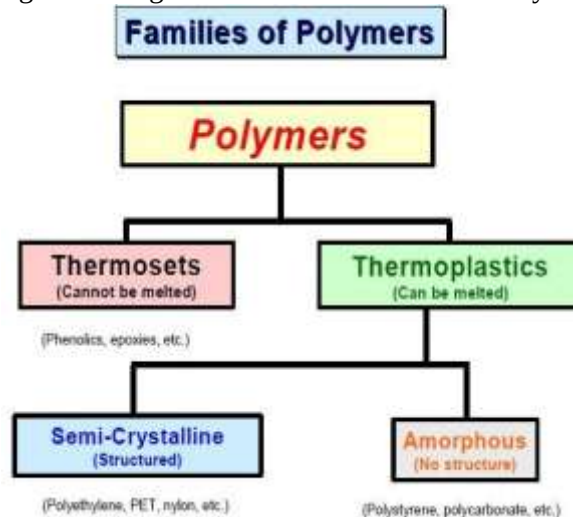


Figure 3 Flow Chart for Polymers

Basic Process: Waste plastic material is grounded and made into a fine powder. Approx. 3-4% of this plastic powder is mixed with the bitumen. This destroyed plastic waste isgoes to acts as a solid restricting specialist. It makes the black-top enduring. At the time of blending, the temperature remains at 55degree Celsius when the plastic is mixed but normally before mixing plastic the blending takes place when temperature reaches at 45.5degree Celsius.

2.1 The Two Main Processes for Manufacturing of Bitumen Mix Roads Using Plastic Waste

2.1.1 Dry Process

In Dry process, the waste plastic is used in high percentage. This is the best process for utilising the high value of plastic waste. The stone aggregate (170°C) is mixed with hot bitumen (160°C) to make a flexible pavement and this mixture is used for road laying. CMP is the another way to carry out the dry process. It is helping to have a uniform coating by better mixing of material and control of temperature.

ADVANTAGES

1. About 10% reduction is occur in the use of bitumen.
2. Use maximum amount of plastic waste.
3. Increase in the performance and strength of the roads. They are not degradable even after the years of construction. Figure 4 provides Flow Chart of Dry Process

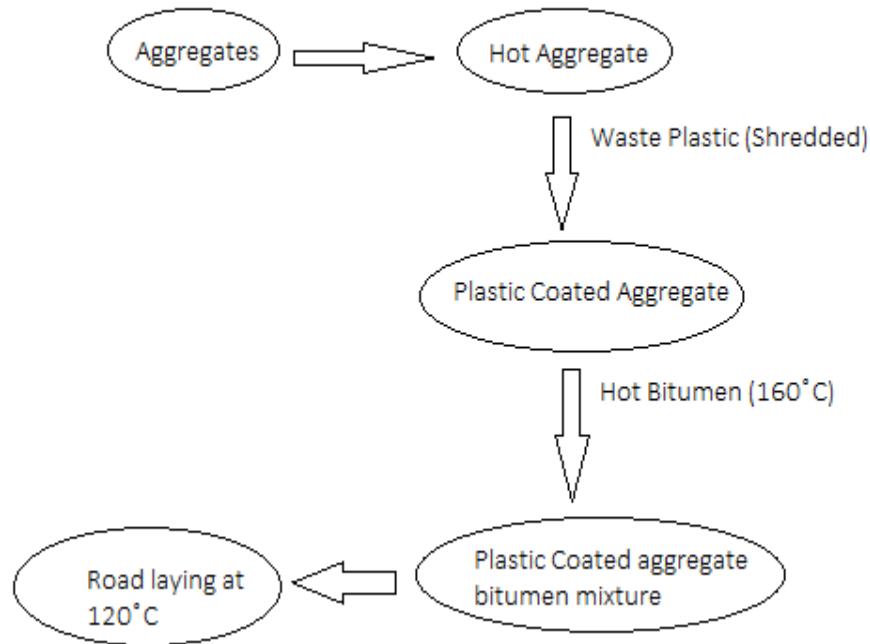


Figure 4 Flow Chart of Dry Process

2.1.2 Wet Process

Firstly, collect the waste plastic and then cut into fine pieces. At 160degree Celsius, the waste plastics are blended by the direct mixing of shredded plastics with the hot bitumen. This process is not commonly used because it requires a bigger plants and the lots of investments. There is difference in viscosities of molten polymer and bitumen which cause the difficulty in mixing. Therefore, Mechanical stirrer is needed in this process. ADVANTAGES: Procedure might be used for reusing of any sort, shape, and size of waste material [10]. Figure 5 shows Flow Chart for Wet Process. Figure 6 shows Plastic Tar making process with the help of pictures. Figure 7 shows Construction of plastic bitumen road.

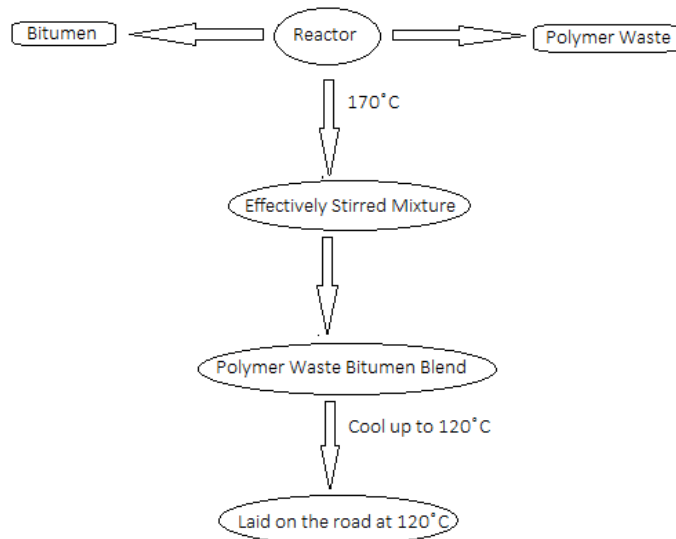


Figure 5 Flow Chart for Wet Process [9]



Figure 6 Plastic Tar making process with the help of pictures



Figure 7 Construction of plastic bitumen road

COMPARISON I between Normal Roads and Plastic Roads

1. At the same time as an ordinary 'highway high-quality' avenue lasts 4 to 5 years it's miles claimed that plastic-bitumen roads can last as long as 10 years.
2. Rainwater will no longer seep via because of the plastic within the tar.
3. The binding assets of plastic makes the road ultimately longer besides giving strength to withstand more loads.
4. The sturdiness of the roads laid out with shredded plastic waste is a good deal greater in comparison with roads with asphalt with the ordinary blend.
5. The maintenance cost of plastic roads is very less in comparison to the conventional method.
6. However, the cost of plastic road construction may be slightly higher as compared to the conventional methods.
7. And the most important comparison is, By the construction of plastic roads we can reduce the non-biodegradable waste.

2.2 Material Cost Comparison for 1km Road

A test that has been successfully managed in many areas, consisting of Kalamassery in Kerala and in Kolkata and Bangalore, may be the reuse of plastic waste to pave roads. The first technological solution, developed by K K Plastic Waste Management Limited, based in Bangalore, involves the use of plastic waste along with bitumen, the ingredient used conventionally to make roads. Bangalore city generates nearly 15 tons of waste plastic bags every day. A plastic waste accumulated around 650 tonnes/ annum. The roads available are about 400km and their annual requirement of plastic waste is more than 600 tons. In this way, the total waste produced can be used for road laying. The lifestyles of the road are improved and therefore the upkeep expenditure is reduced.

Cost of Bitumen per ton is 50000 and to lay the 1Km of normal road the 10 tons of bitumen is required. So the cost of bitumen required per Km is Rs 5,00,000. In Plastic roads, total quantity of bitumen required is 9 tons and the plastic waste required is 1 ton. So the cost of bitumen for 9 tons is Rs 4,50,000 and cost of plastic waste is Rs 5000. So the total cost of plastic and bitumen is Rs 4,55,000. Therefore, total saving is $5,00,000 - 4,55,000 = \text{Rs } 45,000$ per Km . Table 5 provides detail about Cost Comparison. Figure 8 shows Plastic Road Life Cycle.

Table 5 Cost Comparison

Size of the Road	Bitumen Needed	Plastic Needed	Bitumen Saved	Cost Reduced
1Km x 3.75	9 Tonnes	1 Tonne	1 Tonne	Rs 15,000



Figure 8 Plastic Road Life Cycle [23]

3Synthesize the Graphene Nano Sheets from the Waste Plastic Material

Waste plastic control and changing it into value added products is one of the greatest challenges earlier than the scientific network. Hence requiring pressing strategies for plastic solid waste management with the ability to transform the “Waste into Wealth”. Therefore, for the synthesis of graphene Nano sheets the solid plastic waste is one of the outstanding waste that can be used as a forerunner. By making graphene Nano sheets, we not only clean up the environment but also get a new idea to recover the plastic. The synthesis of high quality carbon nano materials from plastic waste is reported by the several studies, where polystyrene (PS), polypropylene (PP), and polyethylene (PE) show their outstanding potential to act as a forerunner for the synthesis of the various forms of CNM’s, such as carbon nano tube’s (CNT’s), carbon nano fibres (CNF’s), and carbon spheres (CS’s). To get the 2-dimesional carbon nano materials, the Nano clays have recently been employed with some solid wastes. The speciality of this process is that, the zero waste remains after the whole process. Therefore, for the wider scope of waste management, the present study motivates towards the process of “Trash to Treasure”. Figure 9 shows Flow chart showing graphene nano sheets and spectral data’s.

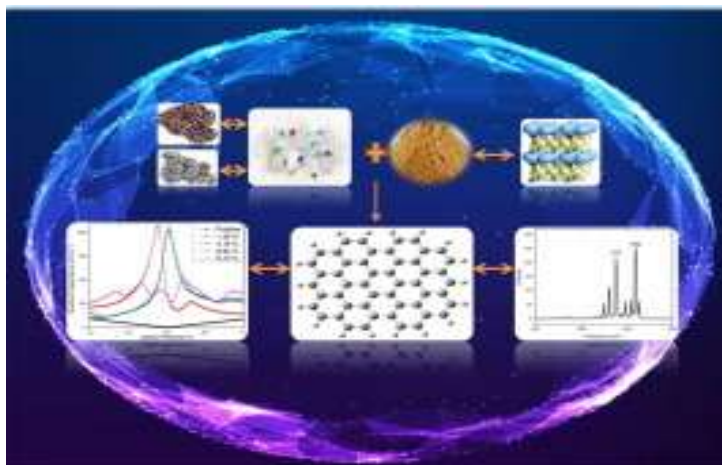


Figure 9 Flow chart showing graphene nano sheets and spectral data's.

3.1 Materials and Method

Firstly, collect the material i.e., waste plastic (about 35 kg; sorted Polypropylene, Polyethylene and Polystyrene) from the local municipality and waste material market. The procedure of making graphene nano sheets from the waste plastic material acquires in several steps.

There's one section which belongs to the slow pyrolysis. It is as much as 400°C with the warming charge of 5°C/min, that's vital put off all hydrocarbon containing oils from the plastic. At this temperature, the nucleation of porous, amorphous and shining black charred residue is caused by slow pyrolysis, which can be gathered by the pyrolysis unit after cooling at room temperature (fig. 10 a). In Secondary reactor, the obtained fine powder after grinding the shapeless residue of the carbon undergoes quick warming rate at 750°C with the warming rate of 10°C/min in dormant atmosphere of N₂(20ml/min), where black coloured graphene Nano sheets obtained (fig. 10b).

3.2 Characterizations

To confirm the success of the amalgamation of graphene nano sheets, numerous spectroscopic and advanced imaging methods were completed. Raman spectroscopy and FTIR spectroscopy is used in this process. We also use some computational analysis for better understanding and the results. Gaussian09 is used for the calculation of DFT (Density Functional Theory).

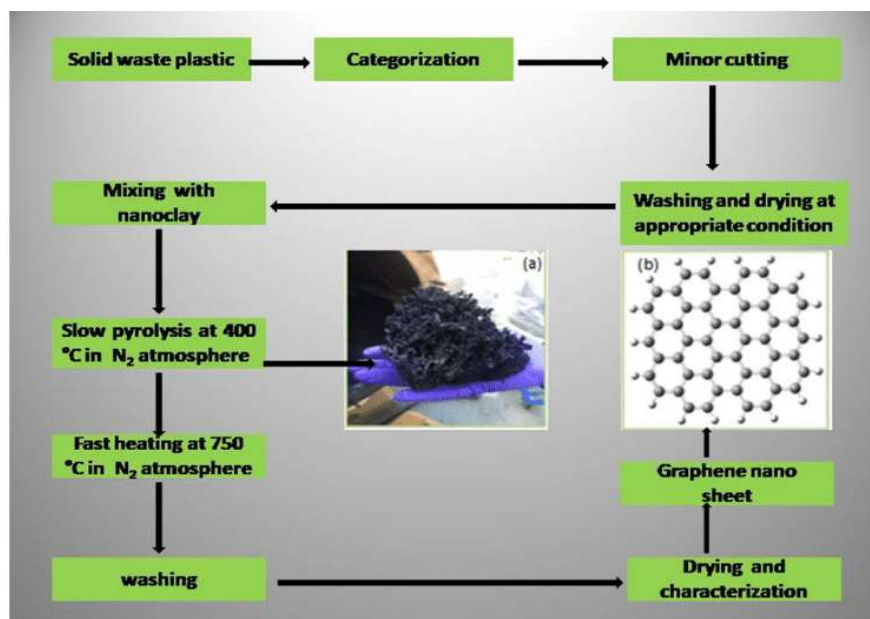


Figure 10 The technique stream outline of creation of graphene Nano sheets, (a) a state of being of carbon build-up acquires from moderate pyrolysis, (b) After quick heating, the skeletal structure of graphene nano sheets [29].

3.2 Results

For the combination of graphene and to survey its condition we can perform the Raman spectroscopy. Inside and the outside if the plan bonding between C – C shows the various active modes by the synthesized sample [30]. The two prominent bands are obtained from the ranges present in the graphene Nano sheets, the one is D-band and the other is G-band. Because of the stretching at sp^3 carbons the D-band is acquired at 1342 cm^{-1} and G-band is at 1594 cm^{-1} due to the stretch occurring at sp^2 carbon (fig. 11a). Two primary sorts of edges: crisscross and armchair are shown by hexagonal structure of graphene (fig. 11d), where the strong D-peak obtain by the main armchair ends are answerable to get flexible dispersing of charge transporter, be that as it may, this impact is fewer mutual for the crisscross ends, however the D-peak intensity for certain examples attributed to the non-consistency and indiscretions in ends structure. Approximately 12 Å , diameter is obtained by raman data of graphene Nano sheet (shown in fig. 11b), wherever the two conspicuous peaks at 1379 cm^{-1} and 1596 cm^{-1} relate to trademark D and G bands, individually. Within the graphene Nano sheets, there are the occurrence of few oxygen covering function groups which are reported by

the FT-IR spectra. Here, the complete weight reduction was 44% of graphene Nano sheets occurred in two phases for example at around the 50 to 400 and 500 to 800 .

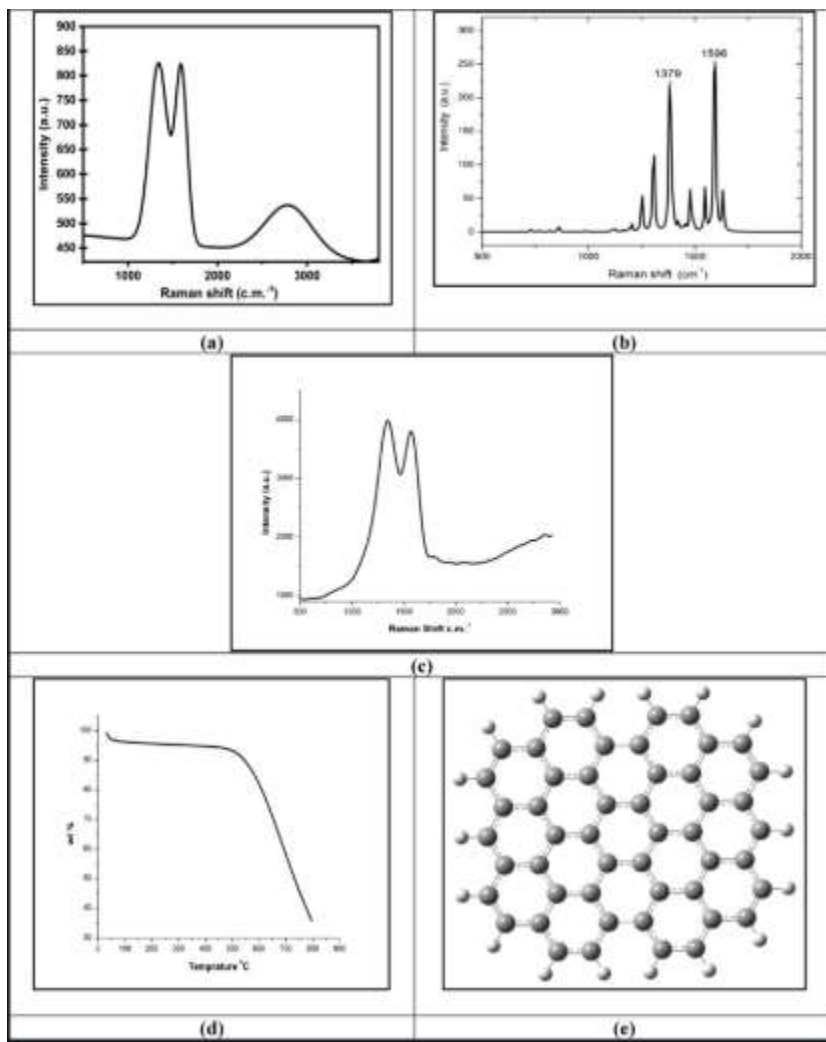


Figure 11 (a)Spectra of graphene Nano sheets by the Raman spectra (Test); (b)Image of Raman spectra from the computational modelling, (c) Raman spectra of black charred residue (d) TGA bend, (e) hexagonal structure of graphene Nano sheets presenting two main types of edges crisscross and armchair

The large scale manufacturing of graphene Nano sheets from solid waste is accounted for by a cost effective method. Raman, FT-IT, TEM, and Energy-dispersive X-ray spectroscopy were executed to affirm the amalgamation of graphene sheets containing the functional group credentials. The synthesized graphene might be utilized as energy storage applications, sun based cells, power devices, drug delivery, Nano composites, and so on.

4 Waste Plastic Oil Production

In pyrolysis, we obtain a mixture of oil similar to crude oil by heating the plastic waste in the absence of oxygen. Further this can be refined into the transportation fuels. Plastic is simply incinerating by the two completely different processes which are Gasification and Pyrolysis.

Polythene is selected because the supply of waste plastic since it accommodates an outstanding percent of the waste plastic produced.

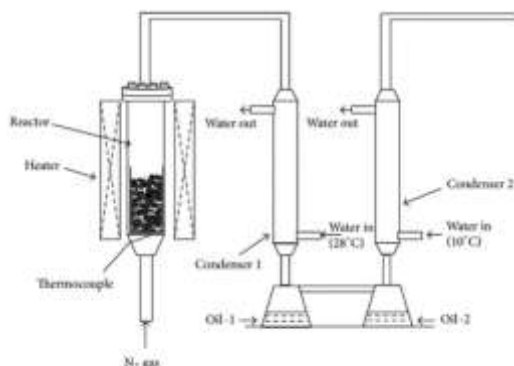


Figure 12 Representation ideal of the trial arrangement [34].

This above setup consists a two oil collectors, two condensers, a steel container, a nitrogen source (inert gas), and a ceramic electric heater (reactor). Maximum 15 kg of plastic is loaded in the reactor. This container is placed inside the heater. The oil acquired from the pyrolysis procedure is can't be straightforwardly utilized as a fuel, because of the presence of certain impurities. The distillation bubble cap tray column is utilized to cleanse the pyrolysis items which decreases the debris and wax content in fuel items. A computerized thermocouple Leybold series 666 209 is utilized to quantify the temperature of every condensate going over the plate. Maximum four number of trays are operated by this process. This is on the basis that in ideal process conditions (580°C), the fourth plate does not have condensation, despite the fact that a condenser unit with cold water ($\pm 5^\circ\text{C}$) is included. Figure 12 shows Representation ideal of the trial arrangement

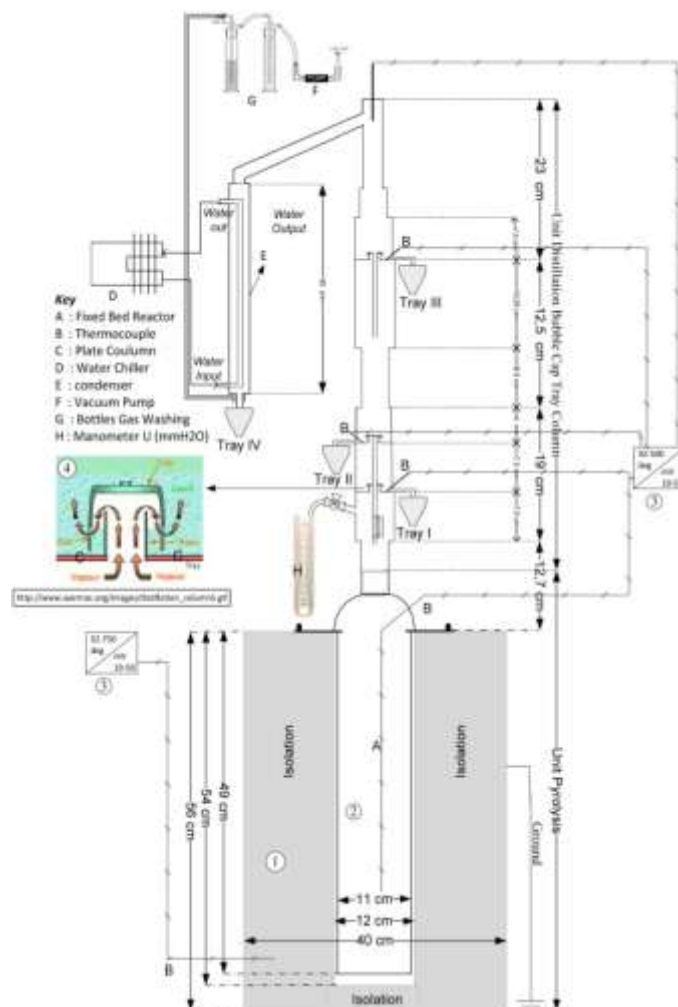


Figure 13 Systematic scheme of thermal cracking pyrolysis coordinated by unit distillation bubble cap tray column.

The fuel oil obtained from the waste plastic material via pyrolysis and unit distillation bubble cap tray column could utilized straightforwardly to the transportation or engine fuel for kerosene and gas type, while diesel fuel is again to meet the requirement of fuel oil. Presently as opposed to thinking of it as a substitute fuel, the sensible use of this system is squander plastic organization incorporates its incentive as a substitute fuel. Figure 13 shows Systematic scheme of thermal cracking pyrolysis coordinated by unit distillation bubble cap tray column.

5 Other Methods

Some other methods to manage the plastic waste are: reuse, recycling, landfills and incineration. Reuse of plastic waste helps to save the time, money, energy, and resources. The process of recycling the plastic waste is to convert the waste material into new items to forestall the waste of helpful materials. The oldest form of waste material treatment is a landfill which is otherwise called a dumping ground site for the removal of waste materials by entombment. Incineration is a process to convert the waste into heat, debris, and vent gas.

5.1 Reuse

It is a middle-man between reduce and recycle. Some would be surprised to see how many opportunities are there to reuse the waste. Without undergoing a physical change, we can use a product for the same or new purpose by the reuse of waste which is known as waste reduction strategy. In this process, energy is not used. Most of the people skip this step and directly go to the recycling, but reusing the plastic is the best process which can reduce the demand for new plastics to be created. Extra tough and multi-trip plastic bundling has become an enormous style in India in current ages, supplanting much fewer strong and single-trip options.

5.2 Recycling

This is the way towards recouping the waste plastic and reprocess again into helpful items. There are several advantages of recycling and re-utilizing of the plastic wastes. Plastic recycling can be complicated by the number of factors, such as the collection of plastic waste, separation of different types of plastic, and cleaning of the waste. Currently the most common method is mechanical recycling which is used to recycle the plastic waste; the term covers its collection, sorting, washing, and grinding. Post-industrial wastes are usually clean, it doesn't have any organic residues and has known compositions. Recycling plastic takes less energy than making plastic from raw materials.

5.3 Landfills

For managing the plastic waste, Landfill is the main traditional approach. But there is one issue in most of the countries in the world i.e. space for landfill. Removal of wastes into landfills infers an irrevocable harm of significant crude resources and vitality. A low density of non-biodegradable plastic waste, makes them unfit for removal in landfills. Solid waste is dumped into the dump yards where waste is either buried or left as it is. In

India, solid waste generation per capita varies from 0.17 kg/individual/day in modest communities to roughly 0.62 kg/individual/day in urban areas. In Landfills, plastic degrades into leeches and these leachates from landfills contain metals, such as lead, mercury, cadmium along with the pesticides, disinfectants, chemicals, pharmaceutical wastes, organic compounds and so on. On the top of the hill, the waste is stored in 5 acres' areas. For the landfilling and dumping, all the inorganic materials are used.

5.4 Incineration

This process discourages the need of landfills of the waste plastics and it tends to be utilized for the recuperation of the energy part of the plastic. The volume of solid waste is diminished by consuming the loss in appropriately structured and working conditions, this process is known as incineration. The burning and the other higher temperature squander treatment frameworks are depicted as "Thermal treatment". The burning of PVC in incineration temperature. Damaging plastic structures by the destruction process which can be done by landfilling or incineration.

6 Conclusion

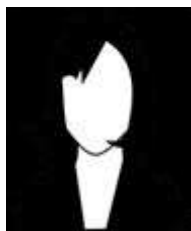
- The collected data shows that we can control the plastic waste by some appropriate methods.
- We can save our environment from this hazardous increase in plastic waste by the above methods.
- This article finds the importance of managing plastic waste.
- The methods discussed in this article are easy, pocket friendly and good for the environment.
- By this article we focus on the problems that occur due to plastic waste and the methodology to control that waste.
- By the use of road construction, we can get a flexible and better durable road.

References

- [1] B.-L. Chen, W. Mhuantong, S.-H. Ho, J.-S. Chang, X.-Q. Zhao, and F.-W. Bai, "Genome sequencing, assembly, and annotation of the self-flocculating microalga *Scenedesmus obliquus* AS-6-11," *BMC Genomics*, vol. 21, no. 1, 2020.
- [2] K. Xu, H. Bao, C. Tang, K. Maliutina, F. Li, and L. Fan, "Engineering hierarchical MOFs-derived Fe-N-C nanostructure with improved oxygen reduction activity for zinc-air battery: the role of

- iron oxide,” *Mater. Today Energy*, vol. 18, 2020.
- [3] H. Qu et al., “Controls on hydrocarbon accumulation by facies and fluid potential in large-scale lacustrine petroliferous basins in compressional settings: A case study of the Mesozoic Ordos Basin, China,” *Mar. Pet. Geol.*, vol. 122, 2020.
- [4] R. S. Park et al., “Advanced Pointing Imaging Camera (APIC) for planetary science and mission opportunities,” *Planet. Space Sci.*, vol. 194, 2020.
- [5] F. Hassanzadeh, N. Daneshvar, F. Shirini, and M. Mamaghani, “Introduction of a new bis-derivative of succinimide (Bis-Su) as a sustainable and efficient basic organo-catalyst for the synthesis of arylidene malononitrile and tetrahydrobenzo[b]pyran derivatives under green conditions,” *Res. Chem. Intermed.*, vol. 46, no. 11, pp. 4971–4984, 2020.
- [6] R. Priyadarshi, B. Kumar, and J.-W. Rhim, “Green and facile synthesis of carboxymethylcellulose/ZnO nanocomposite hydrogels crosslinked with Zn^{2+} ions,” *Int. J. Biol. Macromol.*, vol. 162, pp. 229–235, 2020.
- [7] X. N. Pham, M. B. Nguyen, H. S. Ngo, and H. V Doan, “Highly efficient photocatalytic oxidative desulfurization of dibenzothiophene with sunlight irradiation using green catalyst of Ag@AgBr/Al-SBA-15 derived from natural halloysite,” *J. Ind. Eng. Chem.*, vol. 90, pp. 358–370, 2020.
- [8] L. Hou, Y. Song, F. Lang, Z. Wang, and L. Wang, “Fluorometric determination of Fe^{3+} and polychlorinated benzenes based on Tb^{3+} -pyromellitic acid coordination polymer,” *J. Ind. Eng. Chem.*, vol. 90, pp. 145–151, 2020.
- [9] S. Ward, F. Meng, S. Bunney, K. Diao, and D. Butler, “Animating inter-organisational resilience communication: A participatory social network analysis of water governance in the UK,” *Heliyon*, vol. 6, no. 10, 2020.
- [10] M. Sharma, N. Satyam, and K. R. Reddy, “Strength Enhancement and Lead Immobilization of Sand Using Consortia of Bacteria and Blue-Green Algae,” *J. Hazardous, Toxic, Radioact. Waste*, vol. 24, no. 4, 2020.

Biographies



Gunjan, Department of Chemistry, University Institute of Science, Chandigarh University, Gharuan, Mohali, Punjab.



Ruchi Bharti, Department of Chemistry, University Institute of Science, Chandigarh University, Gharuan, Mohali, Punjab.



Renu Sharma, Department of Chemistry, University Institute of Science, Chandigarh University, Gharuan, Mohali, Punjab.